

## **Tamir assignment**

### **General instructions:**

- **Make sure to read all the instructions of the whole assignment before starting.**
- If there are links here, please open and read them.
- Do not wait to work on this assignment, start working on it whenever you can as it might take more time than you think.
- **I work on these assignments a lot** just for you guys and create them specifically to make your mind and skills expand, so I highly recommend that you really devote your time and effort and do your best to do them properly. It will benefit you a lot.
- Make sure to **enjoy** this exercise. This is the number 1 goal.
- If you feel stuck/lost let me know.

### **Proof of concept of your own game (core gameplay only)**

You will work to create a proof of concept of a web based game.

First choose a game subject/genre. (you can try multiple genres and see what's better for you).

General subjects for games: (these are just some of my ideas. Feel free to choose any game idea)

- Word games (like WOW, Wordle, Crossword puzzle...)
- Side scrolling platformers (like super mario)
- Sports physics games (like head ball, basketball shooting games)
- Card games (like war, uno)
- Endless runners (like doodle jump, flappy bird)
- Classic arcade games (like galaga, breakout)
- Virtual pet games (like tamagotchi)
- 'Cannons shooting balls' games
- 'Snake' type games
- Match3 games

Then think of a simple game that you can create in that genre.

If your game is a 'clone' of another game (most games are), make sure to think of some twist that will make it feel 'your own'. A twist can be a small change in the

game mechanics, a unique powerup that changes something... (feel free to consult me if you don't have an idea).

Use a gen-ai tool to create a banner for your game.

You can try these image creation tools:

- Chatgpt (dall-e) - **free**
- Midjourney - paid, \$10 per month (but amazing)
- Ideogram - **free**
- Leonardo.ai - **free**
- **Mistral chat - free (for [example](#))**

For example, i used this prompt in midjourney

**PC game box cover. Block breaker game, ball, paddle and many bricks, retro, neon**

And this was the resulting image



You can actually use chatgpt to help you create prompts for art. For example i gave chatgpt this prompt:

Help me create a prompt for an art. I want something that will look like the box of a pc game that i am developing. The game is similar to the famous game Breakout and is called 'BlockBuster Adventure'

This created a prompt and i gave the resulting prompt to chatgpt and other gen-ai text to image engines.

Chatgpt (dall-e) created this



So now you have a game idea and a general 'banner'/'box' that makes you feel 'closer' to a real game. That's the point of this image, just to make you feel good 😊

Now you should ask chatgpt or other LLMs (i recommend to use a paid account of chatgpt/claude etc if possible so you can choose the 'strong' llm models) to create you a very simple game:

For example, here is an example chat i had with chatgpt

<https://chat.openai.com/share/67faa949-3c57-413a-bc0b-e47e959c373f>

If you copy paste that code to visual studio like we did in exercise 1 or even in an online html viewer like [this](#) you will see a basic playable game.

So here are prompt formats ideas that you can try and use: (feel free of course to write your own prompts, these are just ideas)

- You are an expert javascript game developer. Create a simple HTML5 flappy bird type game. Just show me the full code. All the code should be in a single file
- You are an expert javascript game developer. Create a simple HTML5 flappy bird type game. Just show me the full code. No explanations and no code placeholders. The game should be in a single file. When the game is over there should be a restart button that restarts the game
- You are an expert javascript game developer. Using pixijs, Create a simple HTML5 flappy bird type game. Each mouse click should cause the bird to fly up. Just show me the full code. No explanations and no code placeholders. Do not use any images. The game should be in a single file. When the game is over there should be a restart button that restarts the game
- You are an expert javascript game developer. Using matter js, Create a simple HTML5 physics game in which the user controls a cannon that shoots balls and the balls should destroy a tower. Each mouse click should shoot a ball and the mouse movement controls the direction of the cannon. Just show me the full code. No explanations and no code placeholders. Do not use any images. The game should be in a single file. When the game is over there should be a restart button that restarts the game

You can try to implement a few different basic games and see what feels better for you.

(fyi, pixi is a popular javascript game engine and matter is a popular javascript physics engine which you can use in your prompts if you want. You can look at demo of Matter [here](#) to understand its possibilities)

Remember again- you do not need to create a full game. Just a working proof of concept that is fun for you.

You can see some of my chats using such prompts [here](#) , [here](#) and [here](#) (You can copy paste the code from my chats here and you will see these are playable stuff).

[Here](#) is another example of match3 game that i created using chatgpt o1 model in a few seconds (only 2 prompts)

[Here](#) is another example, now of a PANG style game that i create using chatgpt o1 using just a few a prompts

[Here](#) is another example of a pachikno style game (similar to Peggle) that i created using chatgpt o1 using just a few prompts. As you can see, there were a few errors in the browser console and i gave those errors to chatgpt and it solved them as well.

Remember- when working with LLMs, if you feel stuck, it is best to create a new chat and not continuing the same long chat (long chats cause more hallucinations and memory issues for the LLMs).

After you implement a basic game, please find some similar games online and try to play them and see what works/doesn't work for you in those games vs your game.

Your game should run for at least 10 seconds.

Remember- this is a proof of concept. You don't have to add levels/ difficulty/ graphics/ music/powerups etc at this stage **BUT** the user must understand your core gameplay idea.

Publish your game like we did in assignment #1,, send it to your friends and family and have people play your game(on a desktop pc and not using a mobile phone) and give you their feedback.

Also, **physically sit next to at least 4** different people while they are playing your game. Look at them while they play it (and even better- take videos), hear what they have to say and write everything down.

Fix/change things in your game until you are happy with the result.

Again, it should only be a proof of concept (but if you want to work on it more and save you time in the final project stage that's completely fine).



**Submission instructions:**

Create a pdf presentation called tamir.pdf file that explains your game and how you solved this exercise including:

- Your game name (of course not critical and you can change it later on)
- Your game banner (of course your game doesn't need to look like it. It's just to give a better feeling and understanding)
- Which similar games do you play(online/phone/ game console etc) and what did you learn/take from them?
- How many people played the game?
- What were their feedbacks?
- What did you do in order to improve the game?
- What would you like to add to the final project?
- What did you learn from this exercise?

(You should add screenshots to the pdf to explain things, for example screenshots of before and after a change that you did)

Create a screen recording video of your game running(up to 5 minutes). There are many free screen recording tools online that you can use for that.

Upload the video to a google drive, create a shared link and make sure anyone can view that link.

Add the link of the video to the pdf.

Add the link of your working game to the pdf.

Make sure that your website works and opens the game properly before the submission.

Send this pdf to me.